Action-conditioned privileged distillation

Forward — Supervision

Shared instruction, robot state, noisy action chunk and flow time

FROZEN MULTI-VIEW TEACHER



Agent



Wrist

VLA backbone

Flow-matching action expert

Layer 10

Final action states

v^t

Teacher velocity

Attention & values

Teacher-native cue extraction

Per-action attention

C_{agent}

C_{wrist}

View-wise values → projection × scale

PCD · cue regression

ACL · Pearson correlation

TRAINABLE SINGLE-VIEW STUDENT



Weak view

VLA backbone

Flow-matching action expert

Layer 10

Final action states

Cue predictors

LN + affine

Predicted cues

$\hat{C}_{agent}$

$\hat{C}_{wrist}$

Gated cue fusion

sg

tanh γ

learned gate

Final action representations

Action output

v^s

Training objective

Ground-truth flow supervision

$$L = L_{\text{flow}} + \lambda_{\text{cue}} L_{\text{cue}} + \lambda_{\text{ACL}} L_{\text{ACL}}$$

sg blocks action-loss gradients through predicted cues.

Deployment: student only

Cue prediction and gate retained